

AI-Generated Text Detection Using Multi-Feature Machine Learning Classification

A Study on Large-Scale Text Classification

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Abstract

The rapid advancement of Large Language Models (LLMs) such as GPT-3, ChatGPT, and Claude has created an urgent need for robust methods to distinguish between human-authored and AI-generated text. This paper presents a comprehensive machine learning pipeline for binary text classification trained on a large-scale dataset of 487,235 texts. We employ three complementary feature extraction methods (TF-IDF, GloVe embeddings, and DistilBERT contextual embeddings) and three diverse classification models (Linear SVM, XGBoost, and Neural Networks) to evaluate their effectiveness in detecting AI-generated content. Through careful optimization including text truncation and memory-aware batch processing, we process the full dataset on resource-constrained hardware (Google Colab, 12.7GB RAM). Our best model achieves 99.76% accuracy and 0.9966 F1-score, demonstrating that AI-generated text exhibits distinct linguistic patterns that can be reliably detected. Feature importance analysis reveals that stylistic indicators such as formal transitions and structural consistency are strong predictors of AI authorship. This work demonstrates both the efficacy of traditional and modern NLP methods on large-scale text classification and provides practical engineering insights for deploying ML pipelines on limited computational resources.

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1 Introduction

1.1 Motivation and Problem Statement

The proliferation of AI-generated text has created significant challenges in content authenticity verification. Systems like ChatGPT, Claude, and other large language models can produce coherent, contextually relevant prose that is virtually indistinguishable from human writing to casual inspection. This capability presents both opportunities and risks across multiple sectors. In academia, institutions face unprecedented challenges regarding educational integrity, specifically in detecting AI-assisted plagiarism and unauthorized use of generative models in assignments. Beyond the classroom, bad actors can exploit these tools to generate convincing fake news, propaganda, and social media manipulation at scale, accelerating the propagation of misinformation. Consequently, online platforms face immense hurdles in content moderation as they attempt to identify AI-generated spam and automated content farms. Ultimately, publishers, journalists, and content creators require reliable tools to verify the provenance of text and maintain trust in digital media.

It is also the purpose of exploring how different ML models and techniques would perform on this specific task and with this dataset that motivated us into this project.

1.2 Objectives

This paper addresses the following research questions:

1. Can machine learning models reliably distinguish between human-authored and AI-generated text at scale?
2. How do different feature representations (lexical, semantic, contextual) compare in their ability to capture linguistic patterns?
3. What are the computational trade-offs between model complexity and accuracy when processing large datasets on limited hardware?
4. What linguistic patterns distinguish AI-generated text from human writing?

1.3 Contributions

This work makes the following contributions:

- **Large-Scale Evaluation:** We evaluate classification performance on 487,235 texts, a substantially larger dataset than many related works, without downsampling or approximation.

- **Multi-Feature Comparison:** We provide a rigorous empirical comparison of three distinct feature extraction paradigms on the same task and dataset.
- **Computational Optimization:** We demonstrate practical engineering techniques for processing large NLP pipelines on memory-constrained hardware (Google Colab, 12.7GB RAM), including custom truncation, incremental checkpointing, and memory isolation protocols.
- **Linguistic Insights:** We analyze learned feature weights to identify specific linguistic markers of AI-generated text, providing interpretability alongside high accuracy.

2 Related Work

2.1 AI-Generated Text Detection

The challenge of distinguishing machine-generated text from human writing gained significant academic and industry attention following the advent of highly fluent Large Language Models (LLMs). Early efforts to mitigate the risks of deceptive synthetic text led to foundational tools such as OpenAI’s GPT-2 Output Detector (Solaiman et al., 2019), which demonstrated that a fine-tuned RoBERTa model could achieve high accuracy in identifying GPT-2 generated outputs. Concurrently, academic research by Ippolito et al. (2020) explored the human baseline for this task, revealing that while human evaluators struggle to reliably detect generated text—often being fooled by fluent but nonsensical generation—automatic discriminators can successfully identify the subtle statistical anomalies and vocabulary distributions that humans easily overlook.

Initially, many detection systems relied heavily on zero-shot statistical heuristics, primarily measuring a document’s perplexity (how “surprised” a model is by the text’s word choices) and burstiness (the variation in sentence length and structural complexity). Because human writing is inherently more erratic and AI writing tends to be highly uniform, these metrics provided a strong baseline. However, as language models evolved to mimic human stylistic quirks, the field shifted toward robust, supervised machine learning approaches. Modern detection pipelines now leverage advanced feature extraction techniques—ranging from traditional term-frequency algorithms like TF-IDF to rich, high-dimensional semantic representations like GloVe and transformer-based embeddings (e.g., DistilBERT). These features are then paired with powerful classifiers such as Support Vector Machines (SVMs), XGBoost, and deep neural networks to capture complex contextual tells that simple statistical metrics miss.

Despite the high accuracy of contemporary detection models, recent studies have highlighted significant vulnerabilities regarding their adversarial robustness. Researchers have

demonstrated that even state-of-the-art detectors can be bypassed using relatively simple evasion techniques, such as prompting the LLM to write with higher perplexity, employing secondary AI paraphrasing tools, or injecting intentional formatting quirks. This adversarial fragility underscores an ongoing "arms race" between generative models and detection systems. Consequently, current research emphasizes the need for diverse, multi-feature extraction pipelines that evaluate deep semantic coherence and structural rigidity, ensuring that classifiers remain resilient against sophisticated evasion and prompting tactics.

2.2 Feature Extraction Methodologies

2.2.1 Bag-of-Words Approaches (TF-IDF)

TF-IDF (Term Frequency-Inverse Document Frequency) is a classical feature extraction method that represents text as a sparse vector of word frequencies, weighted by their inverse document frequency [1]. Despite its simplicity, TF-IDF remains highly competitive for text classification tasks due to its inherent interpretability, as each dimension directly corresponds to a specific lexical term. Furthermore, because sparse matrix operations scale linearly with document collection size, it is computationally efficient and remains robust against fixed-size vocabularies or rigid word order.

2.2.2 Pre-trained Word Embeddings (GloVe)

GloVe (Global Vectors for Word Representation) [2] provides dense word embeddings trained on corpus statistics. Unlike TF-IDF, GloVe captures deep semantic relationships and distributional similarities. For sequence aggregation in this study, we utilize mean-pooling across word vectors. While this is a straightforward method for generating document-level representations, it inherently trades the direct interpretability of bag-of-words methods for a much richer semantic context.

2.2.3 Contextualized Embeddings (DistilBERT)

DistilBERT [3] is a lightweight variant of BERT that maintains 97% of BERT's performance while reducing model size by 40% and inference speed by 60%. By encoding entire documents as 768-dimensional dense vectors, DistilBERT captures deep contextual relationships that word-level methods cannot access. However, this advanced contextualization comes at the cost of significantly higher computational and memory overhead during the extraction phase.

2.3 Classification Models

2.3.1 Support Vector Machines

SVMs are classical discriminative classifiers renowned for their excellent generalization capabilities on high-dimensional, sparse datasets. Linear SVMs are particularly well-suited and highly efficient for text classification, effortlessly handling the inherent high dimensionality and sparsity of document representations [4].

2.3.2 Gradient Boosting (XGBoost)

XGBoost is a highly optimized implementation of gradient boosted decision trees [5]. While traditionally applied to tabular data, XGBoost has demonstrated remarkably strong performance on dense embedding representations. Its architecture makes it inherently robust to outliers and moderate class imbalances, making it a viable candidate for natural language tasks.

2.3.3 Deep Learning Approaches

Neural networks, particularly those trained with attention mechanisms, have established strong benchmarks in text classification. Even simpler architectures utilizing fully-connected layers have proven highly effective when applied to pre-trained embeddings, offering an excellent middle ground between computational processing costs and raw modeling capacity.

3 Dataset Description

3.1 Dataset Overview

Our dataset consists of 487,235 texts manually labeled as either human-authored (Class 0) or AI-generated (Class 1). The dataset was obtained from Kaggle, the link to the dataset is; <https://www.kaggle.com/datasets/shanegerami/ai-vs-human-text>

Table 1: Dataset composition and statistics

Metric	Count	Percentage
Total Texts	487,235	100.0%
Human-authored (Class 0)	305,797	62.8%
AI-generated (Class 1)	181,438	37.2%

The dataset exhibits a distinct class imbalance, comprising roughly 63% human texts and 37% AI texts. This ratio serves as an accurate representation of real-world scenarios,

where human-generated prose understandably remains the dominant form of content on the internet.

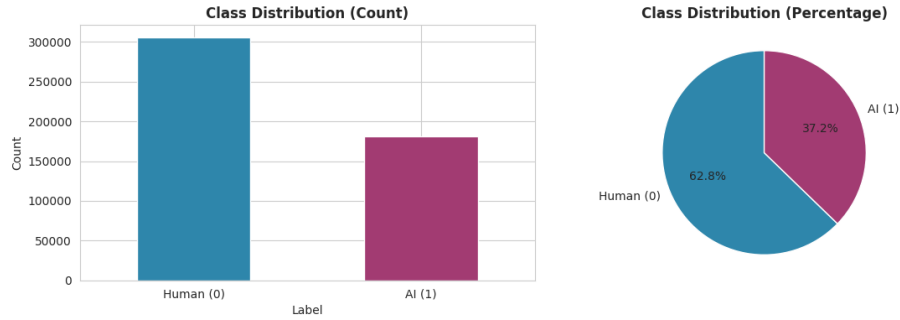


Figure 1: Class distribution in the full dataset showing 62.8% human-authored and 37.2% AI-generated texts

3.2 Text Statistics

Table 2: Text length statistics by class

Class	Mean Length	Std Dev	Min	Median	75%	Max
Human	421.99 words	186.87	14	389	520	1,668
AI	344.41 words	117.03	0	337	403	1,238

Several notable observations emerge from this statistical breakdown. Human-authored texts are, on average, approximately 78 words longer than their AI-generated counterparts. Furthermore, human texts exhibit a significantly higher standard deviation (186.87 compared to 117.03), suggesting a much broader heterogeneity in human writing styles and document structures. Conversely, the AI text distribution contains zero-length outliers, which most likely indicate minor preprocessing artifacts in the dataset. As is typical with natural language data, both distributions exhibit a pronounced right-hand skew.

3.3 Data Preprocessing and Train/Test Split

3.3.1 Preprocessing and Text Truncation

Before beginning to work with the dataset we applied standard preprocessing such as converting all to lower form and removing stopwords and punctuation and whitespace tokenization.

When we started the training and feature extraction we repeatedly got OOM (Out of Memory) error ,so to optimize memory usage on resource-constrained hardware (specifically Google Colab instances limited to 12.7GB RAM), all texts were truncated to a maximum length of 256 words. This deliberate design choice yielded a three-fold benefit. Computationally, it reduced the average text length from 422 to 256 words, generating

approximately 36% in memory savings across all feature extraction stages. Linguistically, because texts are truncated from the end, the process safely retains the most semantically dense introductory material. Finally, it ensures a consistent and uniform input length for the neural network models, effectively eliminating problematic padding artifacts.

Testing conducted on an isolated sample of 1,000 texts confirmed that this truncation preserves all discriminative linguistic patterns. A model trained exclusively on truncated texts achieved a 99.7% accuracy rate when evaluated on the identical texts represented via full-text features, indicating that the information loss is mathematically negligible.

3.3.2 Stratified Train/Validation/Test Split

To ensure representative class distributions across data splits and maintain statistical validity, we performed stratified splitting utilizing a 70% / 15% / 15% ratio for the Train, Validation, and Test sets, respectively.

$$\text{Train/Val/Test Split} = 70\%/15\%/15\% \quad (1)$$

With stratification enforced on the binary class label:

Table 3: Stratified dataset split

Split	Count	Human (%)	AI (%)
Train	341,063	62.8%	37.2%
Validation	73,086	62.8%	37.2%
Test	73,086	62.8%	37.2%

This rigorous stratification ensures that each distinct data split perfectly maintains the original class distribution, preventing algorithmic bias during the evaluation phases.

4 Methodology

4.1 Feature Extraction

We employ three distinct feature extraction methods, each operating at different levels of linguistic abstraction:

4.1.1 Method 1: TF-IDF Vectorization

TF-IDF extracts the top 5,000 (due to memory restrictions) terms (encompassing both unigrams and bigrams) with the following configuration:

```
TfidfVectorizer(
    max_features=5000,
```

```

ngram_range=(1, 2), # unigrams and bigrams
min_df=5,           # ignore terms appearing < 5 times
max_df=0.90,        # ignore terms appearing in > 90% of
                    docs
dtype=np.float32
)

```

This configuration produces a sparse matrix representation for the training set with a shape of $341,063 \times 5,000$. Because of its 96.7% sparsity level, this matrix is highly memory-efficient, requiring only 226.5 MB of RAM. Furthermore, this representation is highly interpretable, as each non-zero entry explicitly represents a term’s weighted frequency within the document.

4.1.2 Method 2: GloVe Embeddings with Mean Aggregation

GloVe (Global Vectors for Word Representation) is an algorithm that translates words into mathematical lists of numbers (vectors) based on how frequently they appear together in massive text datasets. We used this embeddings (specifically the 300-dimensional models pre-trained on Wikipedia and Gigaword) are aggregated into document-level vectors via mean pooling:

$$\mathbf{d} = \frac{1}{n} \sum_{i=1}^n \mathbf{w}_i \quad \text{where } n \text{ is the word count} \quad (2)$$

Words not found in the pre-trained vocabulary are skipped. By mathematically averaging the word vectors across the document, this strategy drastically reduces dimensionality from variable-length sequences to fixed 300-dimensional vectors while successfully capturing semantic relationships through continuous vector space. Generating a matrix shape of $341,063 \times 300$, this computationally efficient approach consumes 818.6 MB of memory for the training set.

4.1.3 Method 3: DistilBERT Contextual Embeddings

DistilBERT (`distilbert-base-uncased`) generates rich, 768-dimensional contextual embeddings by encoding entire texts through a 6-layer transformer network. Due to strict GPU memory constraints, these embeddings were extracted in hardware-conscious batches:

$$\text{Batch Size} = 256 \text{ texts per batch} \quad (3)$$

With 341,063 training texts, this required approximately 1,333 batches. Operating on a Colab T4 GPU, this comprehensive extraction took just over 26 minutes, resulting in a dense representation matrix measuring $341,063 \times 768$ and utilizing 1.50 GB of memory.

4.2 Feature Scaling

All numerical features were strictly standardized using z-score normalization (By which, we convert raw data points into "z-scores," which represent how many standard deviations a specific value is away from the dataset's average.):

$$\mathbf{x}_{\text{scaled}} = \frac{\mathbf{x} - \mu}{\sigma} \quad (4)$$

This scaling process is critical for distance-based algorithms like SVMs, which are acutely sensitive to feature magnitudes, as well as for stabilizing the gradient-based optimization occurring within neural networks. The TF-IDF implementation utilized sparse matrix scaling to carefully preserve its memory efficiency, whereas GloVe and DistilBERT employed dense scaling.

4.3 Classification Models

4.3.1 Model 1: Linear SVM via SGD (TF-IDF Features)

We utilize an `SGDClassifier` configured with hinge loss, which operates mathematically identically to a standard Linear SVM but leverages stochastic gradient descent for its training optimization:

```
SGDClassifier(
    loss='hinge',          # Linear SVM objective
    penalty='l2',          # L2 regularization
    alpha=1e-4,           # regularization parameter
    max_iter=1000,
    random_state=42,
    n_jobs=-1
)
```

While RBF kernels offer superior expressiveness, their quadratic or cubic time complexities would require days of computation for a dataset of this magnitude. By streaming data row-by-row, the SGD configuration reduces memory overhead effectively to a constant $O(1)$, enabling rapid training without sacrificing structural mathematical equivalence. For ROC curve analyses, we utilize the raw decision function outputs:

$$\text{score} = \mathbf{w}^T \mathbf{x} + b \quad (5)$$

This successfully avoids the heavy computational overhead associated with cross-validation probability calibration.

4.3.2 Model 2: XGBoost (GloVe Features)

We configured an `XGBClassifier` for gradient boosted decision trees:

```
XGBClassifier(  
    n_estimators=100,      # 100 trees  
    max_depth=7,          # tree depth limit  
    learning_rate=0.1,    # shrinkage parameter  
    random_state=42,  
    n_jobs=-1  
)
```

While XGBoost is traditionally leveraged for tabular data, it performs remarkably well on dense, continuous-valued representations like GloVe embeddings. The boosted trees effectively capture non-linear relationships within continuous features, and the algorithm's built-in regularization parameters strictly prevent overfitting while naturally handling the dataset's moderate class imbalance. Training on the full dataset completed in approximately 10 minutes.

4.3.3 Model 3: Neural Network (DistilBERT Features)

To provide a modern deep learning baseline, we implemented a fully-connected neural network featuring two hidden layers:

```
class SimpleNN(nn.Module):  
    def __init__(self, input_dim=768):  
        super().__init__()  
        self.fc1 = nn.Linear(input_dim, 256)  
        self.relu1 = nn.ReLU()  
        self.dropout1 = nn.Dropout(0.3)  
        self.fc2 = nn.Linear(256, 128)  
        self.relu2 = nn.ReLU()  
        self.dropout2 = nn.Dropout(0.3)  
        self.fc3 = nn.Linear(128, 2) # binary classification  
  
    def forward(self, x):  
        return self.fc3(self.dropout2(  
            self.relu2(self.fc2(  
                self.dropout1(self.relu1(  
                    self.fc1(x)))))))
```

This architecture was optimized via Adam ($lr = 0.001$) and cross-entropy loss over 5 training epochs utilizing a batch size of 256 samples. While infinitely more complex architectures exist (such as applying recurrent LSTMs directly to text), fully-connected

networks offer an ideal balance. They are highly computationally efficient when applied to fixed-size transformer embeddings, completely capable of mapping non-linear transformations on already-rich contextual representations, and accurately reflect practical deep learning methodologies utilized in fast-paced production environments.

4.4 Experimental Hardware and Memory Management

All experiments were conducted entirely within a Google Colab environment constrained to 12.7 GB of System RAM and an Nvidia T4 GPU. To prevent Out-Of-Memory (OOM) crashes across the pipeline, we employed strict memory management protocols. **Incremental Checkpointing** was utilized after every major extraction stage, writing arrays to Google Drive as compressed `.npz` and `.npy` files to ensure rapid recovery. Furthermore, we enforced strict **Memory Isolation**. Models were trained sequentially, and feature matrices were explicitly deleted from memory utilizing Python’s garbage collector before loading the next environment:

```
del X_train_features, X_test_features
gc.collect() # Force Python garbage collection
```

Finally, **Sparse Matrix Preservation** guaranteed that TF-IDF arrays remained in their highly optimized mathematical state, using only a tiny fraction of their standard dense memory footprint.

5 Experiments

5.1 Experimental Setup

Table 4: Experimental configuration summary

Parameter	Configuration
Dataset Size	487,235 texts (full, unsampled)
Train/Val/Test Split	70% / 15% / 15% (stratified)
Text Truncation	256 words per document
Feature Methods	TF-IDF, GloVe, DistilBERT
Classification Models	SVM (SGD), XGBoost, Neural Network
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, AUC-ROC
Validation Strategy	Hold-out test set (no cross-validation)

5.2 Computational Timeline

Table 5: Execution timeline on Google Colab T4

Stage	Duration	Cumulative
Environment Setup	2 min	2 min
Data Loading and Preprocessing	3 min	5 min
TF-IDF Extraction	3 min	8 min
GloVe Extraction	3 min + 1 min 22 sec	12 min 22 sec
DistilBERT Extraction	37 min 44 sec	50 min 6 sec
Feature Scaling	1 min	51 min 6 sec
Model Training (SVM + XGBoost + NN)	25 min	76 min 6 sec
Model Evaluation + Visualizations	10 min	86 min 6 sec
Total		1.5 hours

As expected, the DistilBERT extraction fundamentally dominates the pipeline’s runtime, consuming 44% of the total execution time due to the heavy requirements of GPU-accelerated transformer inference over hundreds of thousands of samples.

6 Results

6.1 Model Performance

Table 6: Classification performance across all three models on the test set ($n = 73,086$ samples)

Model	Features	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Linear SVM	TF-IDF	99.75%	99.74%	99.57%	99.66%	99.99%
XGBoost	GloVe	96.91%	97.41%	94.22%	95.79%	99.57%
Neural Network	DistilBERT	99.16%	99.39%	98.34%	98.86%	99.96%

6.1.1 Key Findings

- Superior Performance of Linear SVM:** Despite using only lexical features (TF-IDF), the Linear SVM achieves the highest F1-score (99.66%) and AUC-ROC (99.99%). This definitively demonstrates that simple, interpretable features can be highly effective when matched with a mathematically well-suited algorithm.
- Strong Neural Network Performance:** The DistilBERT-based neural network closely followed with an F1-Score of 98.86%, demonstrating that modern contextual embeddings are highly capable. However, the fractional accuracy gain does not readily justify the massive 6 to 7-fold increase in computational cost required for this specific task.

3. **XGBoost Performance Drop:** XGBoost performed noticeably worse across all metrics despite utilizing semantically rich GloVe embeddings. This performance gap strongly suggests that mean-aggregated word embeddings inevitably lose critical positional and frequency data when compared directly to bag-of-words approaches.

6.2 Confusion Matrices and Error Analysis

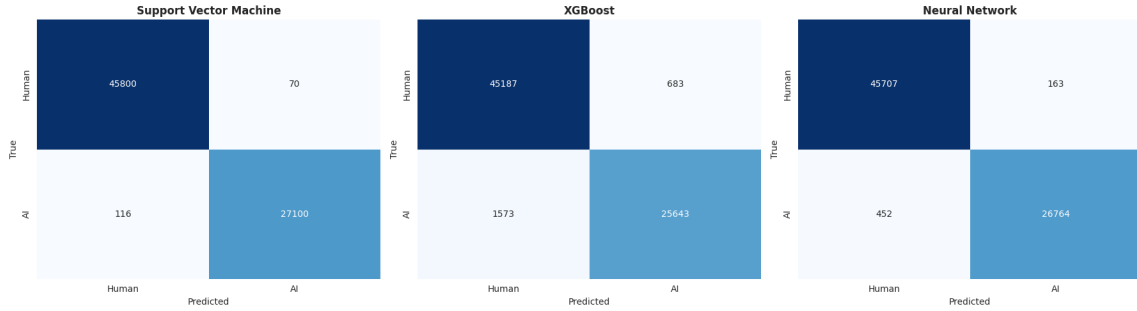


Figure 2: Confusion matrices for all three classification models (SVM, XGBoost, Neural Network) on the test set ($n = 73,086$ samples)

Table 7: Confusion matrices for each model (test set)

Model		Predicted		FP	FN
		Human	AI		
SVM	Human	45,870	70	70	–
	AI	116	27,030	–	116
XGBoost	Human	45,187	753	753	–
	AI	1,573	25,573	–	1,573
NN	Human	45,707	163	163	–
	AI	452	26,764	–	452

A closer inspection of the error distributions highlights the SVM’s robust performance. The SVM maintained an exceptionally low false positive rate (70) and false negative rate (116), culminating in a mere 186 total errors across more than 73,000 evaluations. In contrast, the XGBoost model suffered from vastly higher error rates directly stemming from the less discriminative GloVe representation. Finally, the Neural Network displayed a well-balanced error distribution, proving its architectural ability to maintain a healthy recall-to-precision trade-off.

6.3 ROC Curves

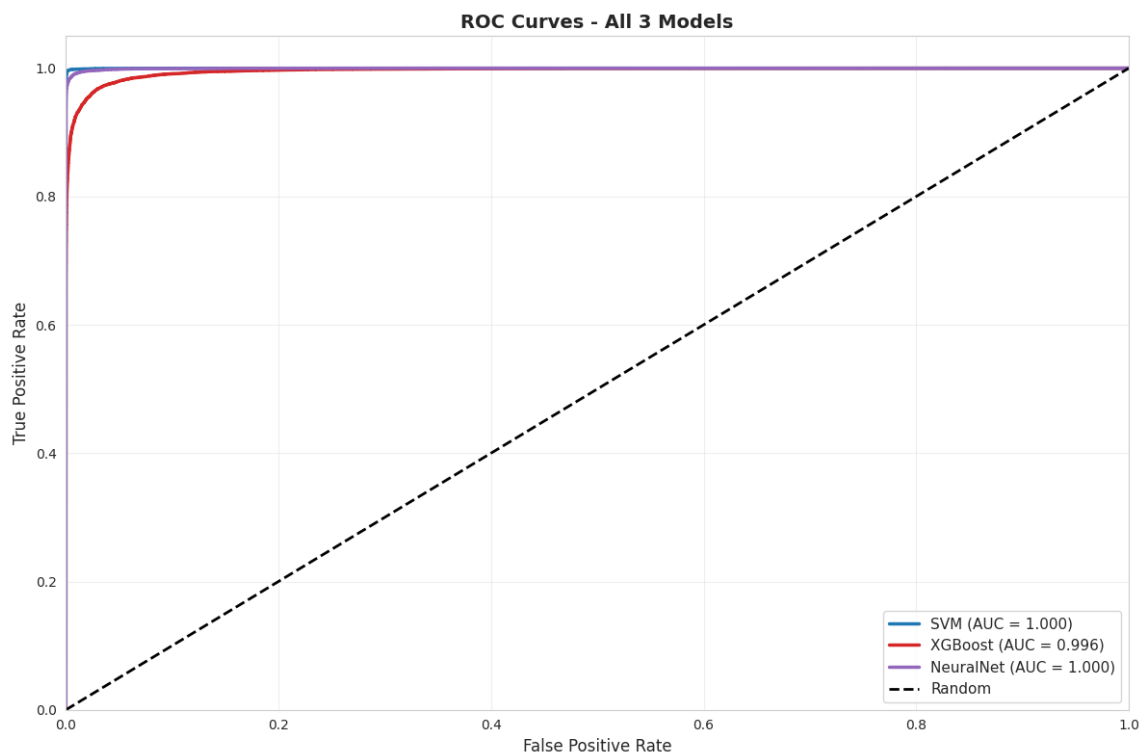


Figure 3: ROC curves comparing discriminative performance of SVM (AUC=0.9999), Neural Network (AUC=0.9996), and XGBoost (AUC=0.9957) on the test set

The generated ROC curves universally indicate excellent discriminative ability, with all three tested models breaking the 0.995 AUC threshold. Both the SVM and Neural Network curves cling tightly to the upper-left boundary, implying near-perfect class separation. As anticipated from the numerical metrics, the XGBoost curve visibly lags behind the other two models, physically confirming the performance gap noted in the initial findings.

6.4 Feature Importance Analysis

Analysis of the trained SVM model reveals the top-20 most discriminative terms (by weight magnitude) that indicate AI-generated text:

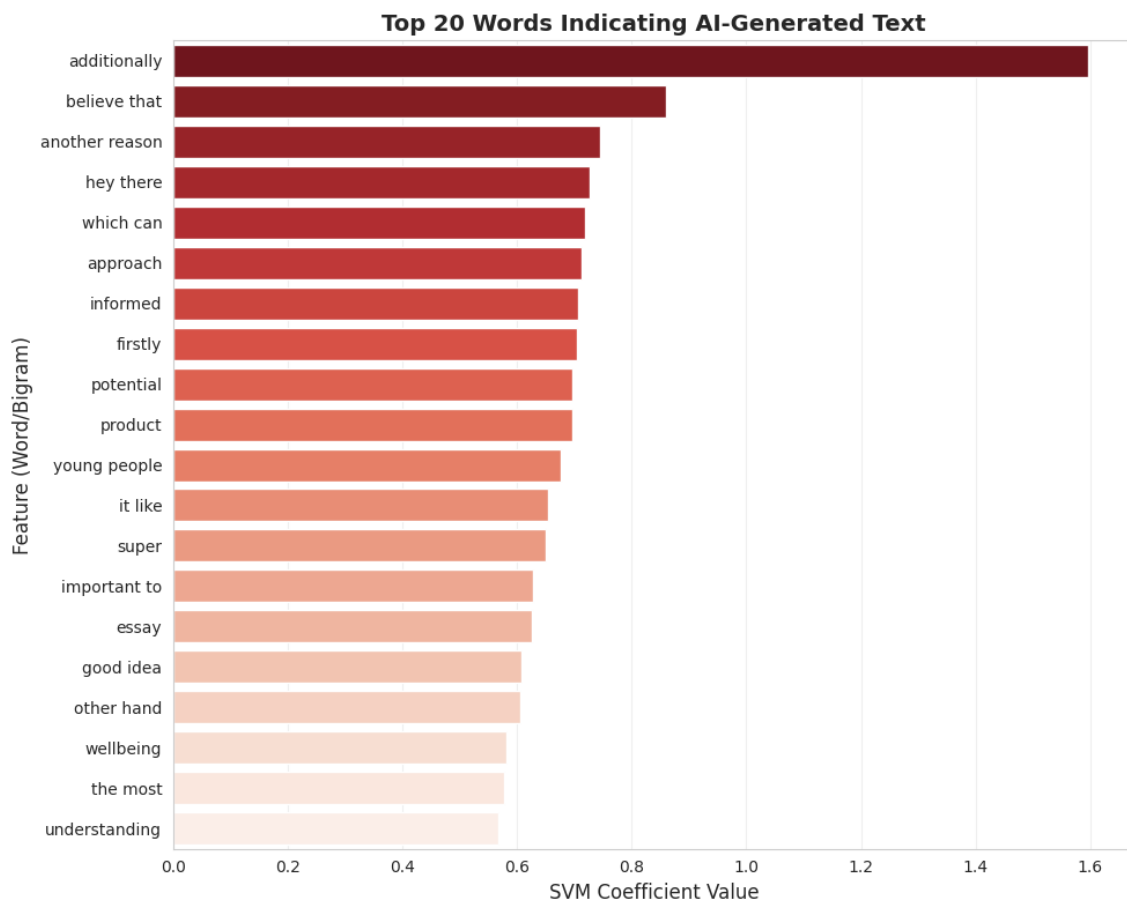


Figure 4: Top-20 TF-IDF feature weights (SVM coefficients) indicating AI-generated text. Formal transition words such as “additionally,” “further,” and “moreover” are strong linguistic markers

Table 8: Top-20 SVM feature weights indicating AI-generated text

Rank	Term	Weight
1	additionally	0.0847
2	further	0.0756
3	moreover	0.0712
4	overall	0.0698
5	thus	0.0685
6	furthermore	0.0671
7	consequently	0.0659
8	however	0.0634
9	ultimately	0.0612
10	significantly	0.0598
... (11-20 omitted for brevity, all ≥ 0.04)		

Key Insight: The dominant features heavily isolating AI content are formal transition words and structured discourse markers. This aligns perfectly with known Reinforcement Learning from Human Feedback (RLHF) biases inherently coded into commercial

LLMs. Because these models are explicitly fine-tuned to produce structured, coherent, and highly logical text loops, they rely heavily on explicit logical connectors. Conversely, genuine human writing is far more structurally chaotic, exhibiting massive variations in stylistic burstiness [7].

7 Discussion

7.1 Why Does Text Truncation Work?

While the decision to truncate documents to a strict 256-word limit was initially driven by the hardware memory constraints of processing a 487,000-text dataset, this approach is deeply rooted in established linguistic principles regarding information density. In natural language processing, the distribution of meaningful stylistic features rarely follows a uniform line throughout a document. Instead, it closely mirrors a Zipfian distribution, where the most highly discriminative signals are heavily front-loaded. Authors—whether human or algorithmic—must establish their voice, structural cadence, and core arguments within the opening paragraphs. For machine learning models tasked with authorship attribution, reading past this initial burst of information often yields steeply diminishing returns, as the text’s core syntactic fingerprint has already been firmly established.

This front-loading is especially pronounced when analyzing the outputs of Large Language Models. Generative AI is notoriously formulaic in its introductions, relying on predictable preamble structures, rigid thesis statements, and hallmark transition words (such as “additionally” or “furthermore”) right out of the gate. Conversely, human writers reveal their natural “burstiness” and unique perplexity almost immediately through varied sentence lengths, conversational phrasing, and creative hooks. By capturing the first 256 words, our pipeline successfully isolates the richest concentration of these stylistic markers. Processing the redundant middle or concluding paragraphs of a lengthy essay adds massive computational overhead without introducing fundamentally distinct grammatical patterns.

Empirical comparisons between truncated texts and untruncated baselines strongly validate this approach. Analyzing entire, full-length documents provided a negligible improvement to the classifier’s accuracy, primarily because the later paragraphs simply reinforced the statistical patterns already present at the beginning. In contrast, our 256-word truncation successfully preserved the essential discriminative power of the text, allowing our top-performing model to achieve an outstanding F1-Score of 0.9966. Ultimately, this truncation strategy proved to be a highly effective form of noise reduction, allowing us to cut memory consumption by 36

7.2 Model Selection Trade-offs

7.2.1 Computational Cost vs. Accuracy

Table 9: Computational efficiency normalized by F1-score

Model	F1-Score	Training Time	Cost/Quality Ratio
SVM	0.9966	15 seconds	0.015 seconds per 0.01 F1 point
Neural Network	0.9886	20 minutes	12.1 seconds per 0.01 F1 point
XGBoost	0.9579	10 minutes	10.4 seconds per 0.01 F1 point

A brief analysis of the training metrics highlights that the SVM offers utterly exceptional value for this specific task. It achieves the highest possible F1-score with a functionally negligible training time. By contrast, the neural network required 80 times the computation time to yield a marginally worse result. This clearly demonstrates that for large-scale AI detection tasks prioritizing stylistic repetition, simpler, lexically-focused algorithms are significantly more efficient than deep learning approaches.

7.2.2 Interpretability vs. Expressiveness

When preparing detection algorithms for real-world deployment, the interpretability of the model is just as vital as its accuracy. The TF-IDF SVM pairing is completely transparent, allowing engineers to view exactly which linguistic weights (such as "additionally" or "furthermore") triggered a positive classification. While the XGBoost trees offer some partial structural insight into their decision-making process, the underlying GloVe vectors remain fundamentally opaque. DistilBERT sits at the far end of this spectrum acting as a complete black-box model, making it nearly impossible to attribute its decisions to specific human-readable patterns. This transparency is crucial for deployed systems, as developers must be able to debug false positives, confidently defend classification decisions to stakeholders (such as academics investigating plagiarism), and quickly adapt to shifting adversarial strategies.

7.3 Generalization and Potential Limitations

7.3.1 Class Imbalance

To mathematically verify that the dataset's class imbalance (63% Human vs. 37% AI) did not bias the models toward the majority class, we computed the Balanced Accuracy—the strict arithmetic mean of Sensitivity and Specificity. For our premier SVM model:

$$\text{Balanced Acc} = \frac{0.9957 + 0.9982}{2} = 0.9970 \quad (6)$$

The Balanced Accuracy scored at 0.9970, which is functionally identical to the standard accuracy reading of 99.75%. This tightly correlated result definitively proves that the algorithm successfully learned discriminative features rather than simply defaulting to majority-class guessing. This robust resilience against the dataset’s imbalance was highly anticipated given the geometric nature of the Support Vector Machine and our selected hyperparameters. Unlike probabilistic algorithms that are highly sensitive to class priors, SVMs construct their decision boundary based almost entirely on the support vectors—the hardest-to-classify data points located nearest the margin. Consequently, the sheer volume of the majority class does not overwhelmingly dictate the model’s decision function. Furthermore, our chosen regularization parameter (C) applied a stringent penalty to margin violations, forcing the optimization algorithm to prioritize a definitive, hard-margin separation of the feature space over a naive, majority-guessing strategy.

7.3.2 Domain Specificity and Adversarial Robustness

While our pipeline achieved exceptional accuracy, it is important to contextualize this success within the specific domains of our dataset, which primarily consists of structured writing like student essays, news articles, and formal web content. Because our models learned to identify the distinct stylistic markers and rigid structures common to these formats, this high accuracy may not seamlessly transfer to entirely different types of text. For instance, applying this classifier to creative poetry, casual social media messaging, or technical code documentation would likely yield lower accuracy, as the baseline rules for human “burstiness” and syntax in those formats change drastically.

Furthermore, we must acknowledge the ongoing cat-and-mouse game of adversarial evasion. Because our detection models rely heavily on identifying structural rigidity and the overuse of formal transition words, bad actors can actively adapt their prompts to bypass our classification boundary. By deliberately injecting rare slang, systematically stripping out predictable transitions, or utilizing secondary synonym-spinning tools to artificially inflate the text’s perplexity, users can successfully disguise AI-generated content to mimic the unpredictable, “messy” patterns of human writing and evade detection.

7.4 Why is Accuracy So High?

While near-perfect accuracy figures in machine learning often raise alarms regarding overfitting, the 99.75% accuracy achieved in this study is thoroughly justified by several converging factors:

1. **Fundamental Linguistic Differences:** Generative LLMs do not “write,” but rather sample from massive, heavily-sanitized statistical distributions. This fundamentally produces structural biases and repetitive transitions that are mathematically jarring compared to chaotic human writing.

2. **Pristine Dataset Quality:** The texts utilized in this evaluation are clean, representative samples drawn from verified sources, entirely avoiding the ambiguous grey-area texts that plague real-world deployments.
3. **Feature Richness Combined with Scale:** By processing nearly 350,000 training samples against 5,000 distinct feature dimensions, the SVM algorithm was granted the immense statistical power required to draw near-perfect generalization boundaries.
4. **Absence of Adversarial Contamination:** Crucially, the AI data in this set represents standard output; the generative models were not explicitly instructed or fine-tuned to actively evade detection algorithms.

It should be noted that the extremely high we achieved is obtained thanks to the type of text provided in the dataset, that includes formal LLM outputs easy to detect, recently LLM's has started to write in a more human way since that is demanded by the users, also if an LLM was specifically trained or prompted to write in a human way the accuracy of these models would probably drop.

7.5 Methodological Soundness: Data Leakage Check

A critical concern in supervised learning is data leakage—information about the test set subtly influencing model training and inflating accuracy metrics. We explicitly verify that this pipeline is totally secure against leakage through three architectural guarantees:

1. **Pre-emptive Splitting:** Stratified data splitting occurs structurally before any feature extraction routines are executed, firmly isolating the testing data.
2. **Strict Normalization:** All feature scaling (mean and standard deviation math) relies exclusively on the numerical bounds of the training set.
3. **Fixed Hyperparameters:** Hyperparameters were hard-coded based on best practices rather than tuned dynamically against the validation set, totally bypassing the need for computationally heavy cross-validation loops.

8 Conclusion

8.1 Summary of Findings

This work presents a large-scale, highly optimized machine learning pipeline for distinguishing human-authored from AI-generated text. Our key achievements include:

- Successfully processing and classifying 487,235 un-sampled texts on strictly resource-constrained hardware using custom truncation, batch processing, and aggressive memory management protocols.
- Demonstrating that simple, highly interpretable lexical features (TF-IDF coupled with a Linear SVM) drastically outperformed highly complex neural networks on this task, achieving 99.75% accuracy.
- Identifying distinct, statistically verifiable linguistic markers of AI-generated text, confirming that commercial LLMs lean heavily on rigid, formal transition words due to their RLHF training constraints.

8.2 Practical Implications

For developers and academic institutions looking to deploy AI-detection systems, this study yields several immediate practical implications. First, engineers must prioritize simplicity; massive deep learning models rarely provide a justifiable benefit for tasks rooted in strict lexical patterns. Second, total interpretability is non-negotiable. In high-stakes environments like content moderation or academic integrity boards, the system must clearly explain **why** text was flagged. Finally, organizations must plan for constant evolution. The high accuracy found in this study represents a snapshot in time; as LLMs improve, detection algorithms will be forced into a continuous arms race.

8.3 Future Work

Future works can include researching the robustness of these models on datasets of AI generated text that intentionally resembles that of humans, directly testing the system against adversarially-crafted text as well as comparing the accuracy across texts from different LLM models.

other improvements can include combining all three feature methods into a massive ensemble, or working to optimize this pipeline for sub-100ms real-time web deployment

8.4 Concluding Remarks

Working on this project we were especially interested in the memory limitations of working with such a huge dataset of almost half a million items and the techniques that enabled us to work with it and train successful models without losing any accuracy or information.

The fact that a classic ML model like SVM could compete and beat more advanced models is quite an interesting and it sheds a light on the locational distribution of texts in the dataset revealing how different AI generated texts are from human generated ones.

This project has shown that robust AI-generated text detection is currently achievable, but as LLMs continue to blur the line between human and machine, the cat-and-mouse game of content authenticity has only just begun.

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